A uniqueness set which is not sampling and whose complement is not a λ-set

Candidate full counterexample packet for arXiv:2511.17343

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Abstract

Under exactly the graph assumptions stated by Giannoni, there is a positive bandwidth ω, an infinite-dimensional Paley–Wiener space $\mathrm{PW}_\omega(G)$, and a uniqueness set W whose complement is not a λ-set. In fact, W is not a sampling set and $(\mathrm{PW}_\omega(G), \|\cdot\|_W)$ is incomplete. The graph is the disjoint union of an eight-cycle and finite 10-regular Ramanujan expanders of unbounded order. The proof reduces to exact finite-graph spectral calculations and one explicit family of test functions.

Status and scope. This is a candidate full affirmative answer to the existence question following source Theorem 3.1, together with a negative answer to the completeness issue raised after source Theorem 2.8. The source assumes countably many vertices, no isolated vertices, simple undirected uniformly weighted edges, and finite maximum degree; it does not assume that the graph is connected. The example uses this stated generality. A connected version is not claimed.

1 Construction and spectral calculation

For a regular graph of degree d, the normalized Laplacian used in the source is

$$\mathcal{L} = I - \frac{1}{d}A,$$

where A is the adjacency operator. Let $(X_n)_{n\geq 1}$ be connected 10-regular bipartite Ramanujan graphs of orders N_n tending to infinity. Passing to a subsequence, assume $N_n \geq n^4$. Such an infinite family exists by Marcus–Spielman–Srivastava [2]. Put

$$G = C_8 \ \sqcup \bigsqcup_{n\geq 1} X_n, \qquad \omega = 1 - \frac{1}{\sqrt{2}} > 0.$$

This is a simple, uniformly weighted, countable graph without isolated vertices and with maximum degree 10.

The adjacency eigenvalues of C_8 are $2\cos(2\pi k/8)$, $0 \leq k < 8$. Hence the normalized-Laplacian eigenvalues are

$$1 - \cos(2\pi k/8),$$

so ω is an eigenvalue of multiplicity two and $\dim \mathrm{PW}_\omega(C_8) = 3$. In particular $\omega \in \sigma(\mathcal{L}_G)$, as required in the source definition of uniqueness set.

On every X_n, the constant functions form the zero eigenspace. Every nontrivial adjacency eigenvalue μ satisfies $|\mu| \leq 2\sqrt{10-1} = 6$; therefore every nonzero normalized-Laplacian eigenvalue is at least

$$1 - \frac{6}{10} = \frac{2}{5} \ > \ 1 - \frac{1}{\sqrt{2}} = \omega.$$

The spectral projection of a Hilbert direct sum is the direct sum of the component spectral projections. Consequently

$$\mathrm{PW}_\omega(G) = \mathrm{PW}_\omega(C_8) \oplus \bigoplus_{n\geq 1} \mathrm{span}\{\mathbf{1}_{X_n}\}. \tag{1}$$

Thus this Paley–Wiener space is infinite-dimensional. Every one of its elements has a unique representation

$$f = g \oplus \bigoplus_{n\geq 1} c_n \mathbf{1}_{X_n}, \qquad g \in \mathrm{PW}_\omega(C_8), \qquad \|g\|_2^2 + \sum_{n\geq 1} N_n |c_n|^2 < \infty. \tag{2}$$

2 The full counterexample

Choose one vertex $x_n \in X_n$ for each n, and define

$$W = V(C_8) \cup \{x_n : n \geq 1\}, \qquad S = V(G) \setminus W.$$

Theorem 1. *For G, ω, W, S above:*

1. *W is a uniqueness set for $\mathrm{PW}_\omega(G)$;*
2. *W is not a sampling set;*
3. *$(\mathrm{PW}_\omega(G), \|\cdot\|_W)$ is not complete;*
4. *S is not a λ-set for any finite λ.*

Proof. For f as in (2), restriction to W gives all values of g and the single value $c_n = f(x_n)$ on each X_n. Hence $f|_W = 0$ implies $g = 0$ and every $c_n = 0$, proving uniqueness. Moreover, the two norms have the exact forms

$$\|f\|_2^2 = \|g\|_2^2 + \sum_{n\geq 1} N_n |c_n|^2, \tag{3}$$

$$\|f\|_W^2 = \|g\|_2^2 + \sum_{n\geq 1} |c_n|^2. \tag{4}$$

Let $h_n = N_n^{-1/2} \mathbf{1}_{X_n}$, extended by zero off X_n. Then $h_n \in \mathrm{PW}_\omega(G)$, $\|h_n\|_2 = 1$, and $\|h_n\|_W = N_n^{-1/2} \to 0$. No uniform lower frame inequality is possible, so W is not sampling.

For incompleteness, let

$$f^{(m)} = \bigoplus_{n=1}^{m} \frac{1}{n} \mathbf{1}_{X_n}.$$

Every $f^{(m)}$ lies in $\mathrm{PW}_\omega(G)$, and (4) shows that $(f^{(m)})$ is Cauchy in $\|\cdot\|_W$. If it converged in that norm to some $f \in \mathrm{PW}_\omega(G)$, then coordinate convergence at x_n would force the coefficient of $\mathbf{1}_{X_n}$ in f to be $1/n$ for every n. But then (3) and $N_n \geq n^4$ give

$$\|f\|_2^2 \geq \sum_{n\geq 1} \frac{N_n}{n^2} \geq \sum_{n\geq 1} n^2 = \infty,$$

a contradiction.

It remains to test the λ-set inequality directly. Let φ_n be one on $X_n \setminus \{x_n\}$ and zero everywhere else. It is supported in S and

$$\|\varphi_n\|_2^2 = N_n - 1.$$

Because X_n is 10-regular, $\mathcal{L}\varphi_n$ equals -1 at x_n, equals $1/10$ at each of its ten neighbors, and is zero at every other vertex. Therefore

$$\|\mathcal{L}\varphi_n\|_2^2 = 1 + 10\left(\frac{1}{10}\right)^2 = \frac{11}{10}.$$

The ratio $\|\varphi_n\|_2 / \|\mathcal{L}\varphi_n\|_2$ tends to infinity, so no finite Poincaré constant λ works on S. $\square$

What the example resolves. The paper asks whether uniqueness sets not arising as complements of λ-sets exist. Theorem 1 supplies one at a genuine positive spectral bandwidth. It simultaneously shows why uniqueness alone does not imply stable sampling: injectivity of the restriction map does not make its range closed. Equations (3)–(4) identify the defect exactly as the missing component-size weights.

3 Source evidence

The source first identifies completeness of the uniqueness norm as unknown (PDF page 8):

> set, this means that if $f\big|_U \equiv 0$, then $f \equiv 0$.
>
> Remark 2 is what is exploited in [12] to prove Theorem 2.8, by using the closed graph theorem applied to the identity operator between $(PW_\omega(G), \|\cdot\|_2)$ and $(PW_\omega(G), \|\cdot\|_U)$. However, this theorem only works for Banach spaces, and we do not know whether or not $(PW_\omega(G), \|\cdot\|_U)$ is complete. This is presumably the reason why the statement of Theorem 2.8 is corrected in the Erratum [11], by adding the crucial assumption that the space $PW_\omega(G)$ must be finite-dimensional. This assumption makes the proof trivial since all the norms defined on a finite-dimensional vector space are equivalent. However, this also makes the results of this theorem far less interesting: the most studied graphs in harmonic analysis (like homogeneous trees - see e.g. [4]- and $\mathbb{Z}^n$-lattices) all have infinite-dimensional Paley–Wiener spaces. Thus, our goal must be finding a sampling theorem that would allow for robust reconstruction in the infinite-dimensional scenario. In the next section, we are going

It then explicitly asks whether uniqueness sets which are not complements of λ-sets exist (PDF page 10):

> Hence
> $$\|f\|_2 \leq \frac{1 + \lambda\omega_{\max}}{1 - \lambda\omega}\|f\|_W$$
> and since $\lambda\omega < 1$, this concludes our proof. □
>
> Observe that this theorem is a stronger version of Theorem 2.6: along with showing that the complement of a λ-set is a uniqueness set, it also shows that it is a sampling set.
>
> This theorem solves our problem only partially, since it provides a proof for norm equivalence only if W is the complement of a λ-set, and we do not know if the equivalence holds whenever W is a uniqueness set which is not the complement of a λ-set. Indeed, we do not even know whether such uniqueness sets exist. However,

References

[1] F. Giannoni, *Sampling on Paley–Wiener spaces on graphs, with particular focus on the infinite-dimensional case*, arXiv:2511.17343v6 (2026), especially Theorems 2.8, 3.1, 3.2 and the intervening remarks.

[2] A. W. Marcus, D. A. Spielman, and N. Srivastava, *Interlacing Families I: Bipartite Ramanujan Graphs of All Degrees*, Ann. of Math. **182** (2015), 307–325; arXiv:1304.4132. They prove the existence of infinite families of regular bipartite Ramanujan graphs of every degree greater than two.